Technical Overview: Retrieval-Augmented Generation & Modern Vector Architectures

1. Introduction to Artificial Intelligence & Language Models
Artificial Intelligence (AI) represents a foundational shift in computer science, focusing on building systems capable of executing tasks that traditionally required human intelligence. Within this domain, Machine Learning (ML) empowers systems to learn patterns directly from empirical data without being explicitly programmed.

Deep Learning, a specialized subfield of machine learning, relies on artificial neural networks with multiple hidden layers to process unstructured high-dimensional data, such as images, audio, and raw text. Recent advancements in deep neural network architectures—most notably the Transformer architecture—have given rise to Large Language Models (LLMs). These models possess vast parametric knowledge acquired through pre-training on massive multi-terabyte text corpora.

2. The Limits of Parametric Knowledge & The RAG Framework
While Large Language Models exhibit remarkable generative capabilities, they suffer from fundamental limitations:
• Knowledge Cutoffs: An LLM's knowledge is frozen at the moment its training process completes.
• Hallucinations: Models may generate plausible but incorrect information.
• High Fine-Tuning Costs: Updating model weights is computationally expensive.

Retrieval-Augmented Generation (RAG) bridges external retrieval databases with neural generation by retrieving relevant documents at query time and supplying them as context.

3. The Core Mechanics of Vector Search & Storage
A. **Document Ingestion and Text Chunking**: Documents are parsed and split into chunks using fixed-size or semantic chunking.
B. **Vector Embeddings**: Each chunk is converted into a dense embedding vector (e.g., 384 or 1536 dimensions).
C. **Vector Databases & Metadata Indexing**: Vectors are stored in databases such as Qdrant, ChromaDB, or Pinecone with metadata including source, chunk index, and timestamps.
D. **Similarity Search & Retrieval**: User queries are embedded and matched using cosine similarity, dot product, or Euclidean distance to retrieve the top-k relevant chunks.

4. Evaluation Metrics for Retrieval Systems
• Precision@k: Fraction of retrieved top-k chunks that are relevant.
• Recall@k: Fraction of all relevant chunks retrieved in the top-k.
• Faithfulness / Groundedness: Measures whether generated answers remain supported by retrieved evidence.